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論文題目	Simulation-Based Rash Behavior Prediction of Surrounding Vehicles Using Multi-Modal Sensor Fusion		

1. Introduction

Autonomous vehicles (AVs) are classified by the Society of Automotive Engineers (SAE J3016) into six levels of driving automation, from Level 0 (no automation) to Level 5 (full automation) [1]. Industry forecasts that Level 4 AV deployment, currently around 1%, will reach only about 5% by 2030, meaning that for the foreseeable future AVs must operate in a hybrid traffic environment shared with human-driven vehicles [2]. The transition from Level 2 to Level 3 and above requires an AV not merely to track the trajectories of nearby vehicles but to anticipate anomalous or unsafe behavior before a critical hand-over or collision-avoidance response is triggered.

This work defines rash driving as a sudden, high-risk maneuver, including an abrupt lane cut-in or emergency brake, performed by a third-party vehicle and posing an immediate collision risk to the ego vehicle, distinct from the “aggressive driving” construct used in the existing literature, which characterizes a prolonged behavioral disposition of the monitored vehicle itself. Existing aggressive driving detection research is almost exclusively ego-centric, scoring a vehicle’s own maneuvers from its own IMU, GPS, and steering data, and provides no information about the behavior of surrounding traffic. This dissertation instead proposes reactive threat anticipation: predicting rash maneuvers in nearby vehicles using only sensor data observable from the ego AV itself, including six-view surround RGB cameras, six-view semantic segmentation, 360° LiDAR rendered as a Bird’s-Eye View (BEV), and a kinematic stream describing ego and nearby-vehicle position, speed, and heading.

The objectives of this work are to (1) construct a purpose-built, frame-precise multi-modal dataset for third-party rash-event prediction; (2) establish whether kinematic data alone is sufficient for this task; (3) design and evaluate a multi-modal fusion architecture combining spatial, visual, and kinematic information through a temporal model; (4) evaluate generalization to unseen road environments; and (5) assess computational feasibility for constrained onboard AV hardware.

2. Dataset Construction

The dataset is constructed in MetaDrive 0.4.3, a lightweight, CPU-only, Panda3D-based simulator that procedurally generates a structurally distinct road topology for every episode as shown in Fig. 1, in contrast with CARLA’s twelve fixed maps [3], [4]. Procedural generation is essential to this work’s generalization claim: as all visual features pass through a frozen, ImageNet-pretrained MobileNetV3-Small backbone, rendering photorealism does not affect feature quality, while unique per-episode topology prevents the model from learning map-specific rather than general behavioral cues.

Each 60-second episode runs at 30 FPS; non-player vehicles are scripted to trigger a lane cut-in or emergency brake at a randomized time, and the recorder captures a sliding window of 3 s pre-event and 2 s post-event frames at 10 FPS (100 ms resolution). All four modalities, including RGB, semantic segmentation, LiDAR BEV, and a 19-dimensional

kinematic vector describing the ego vehicle and up to three nearby vehicles, are captured synchronously, and a binary label is assigned to every frame in the captured window.

Fifty episodes were collected, yielding 68 event clips and 17,559 frames (83.9% rash, 16.1% normal). The split is performed at the episode level, training (57 clips), validation {3, 6, 45, 46}, and test {9, 10, 47, 49}, so that no road layout, NPC trajectory, or background scene is shared across sets, which is essential for a genuine test of generalization. A calibration defect in MetaDrive’s native LiDAR-to-point-cloud conversion (swapped intrinsics and a transposed depth buffer) was identified and corrected by bypassing the built-in routine and back-projecting the raw depth texture directly, without which the BEV grids carry no genuine spatial signal.



Fig.1 Surround camera images collected through RGB camera modality.

3. Proposed Architecture: Early Fusion V2

The proposed model processes four parallel per-frame branches. A kinematic branch projects the 19-dimensional motion vector to a 32-dimensional embedding through a small trainable head. RGB and semantic branches each pass six camera views through a shared, frozen MobileNetV3-Small backbone, mean-pool across views, and project to 64 and 32 dimensions respectively. A BEV branch passes the $64 \times 64 \times 3$ occupancy/intensity/height grid through a three-layer CNN trained from scratch (~89k parameters) to a 64-dimensional embedding. The four embeddings are concatenated into a 192-dimensional per-frame vector; the resulting ten-frame sequence (one second of context) is passed through a two-layer Bidirectional GRU (hidden size 96, dropout 0.5), and the final hidden state is mapped through a classification head (Linear-ReLU-Linear-Sigmoid) to a calibrated rash-event probability. The full model totals 492,193 parameters.

Feature-level (early) fusion is used rather than late fusion so that the GRU can learn cross-modal temporal correlations — for example, relating a change in kinematic closing distance to a simultaneous shift in BEV occupancy. Training uses weighted binary cross-entropy to address the 83.9%/16.1% class imbalance, with a cosine-annealed learning rate over a maximum of 40 epochs and early stopping (patience 12, which is a critical threshold for preventing overfitting during model training).

The two preceding iterations were designed, not adopted from prior literature. As shown in Table 1, Kinematic LSTM uses only the 19-dimensional motion vector through a two-layer BiGRU (hidden 64, ~184k parameters), testing whether motion alone is sufficient. Early Fusion V1 augments this with raw MobileNetV3 RGB features and a 9-dimensional statistical summary of the LiDAR point cloud, in place of the full spatial BEV grid used in V2; its weaker generalization is what motivated replacing the LiDAR summary with a trainable BEV-CNN branch in V2.

4. Experimental Results

4.1. Performances of Different Models

All models are implemented in PyTorch 2.4.1 and trained on CPU hardware only; visual features and BEV grids are pre-cached (46 minutes for all 17,559 frames), and float16 storage keeps memory usage to ~514 MB. Table 2 summarizes held-out test-set performance at each model’s validation-calibrated decision threshold.

Table 1: Model comparison summary

Model	Key Modalities/Components	Validation F1	Test F1
Kinematic LSTM	Kinematics only	0.9275	0.9450
Early Fusion V1	+ RGB (MobileNetV3) + LiDAR statistics	0.9265	0.7657
Early Fusion V2 (Proposed)	+ LiDAR BEV CNN + Semantic Segmentation	0.9455	0.9003

Table 2. Held-out test-set performance of the three models

Model	F1	AUC	Prec.	Rec.
Kinematic LSTM	0.9450	0.9088	0.9347	0.9556
Early Fusion V1	0.7657	0.8927	—	—
Early Fusion V2 (Proposed)	0.9003	0.8976	0.9368	0.8665

As shown in Table 2, Kinematic LSTM alone achieves a strong test F1 of 0.9450, indicating that relative motion — closing speed, distance, and heading — already carries most of the predictive signal for rash-event anticipation. Early Fusion V1 achieves a validation F1 (0.9265) statistically indistinguishable from the kinematic baseline, but its test F1 collapses to 0.7657, which is an important negative finding showing that naively appended visual features and a scalar LiDAR summary discard the spatial structure needed to generalize and can introduce noise rather than signal.

Early Fusion V2 resolves this by replacing the scalar LiDAR summary with the full CNN-processed BEV grid and adding a dedicated semantic branch. It achieves the best validation F1 of all three models (0.9455) and a held-out test F1 of 0.9003 (AUROC 0.8976, precision 0.9368, recall 0.8665) on episodes sharing no road layout with training, which is an evidence of genuine generalization rather than memorization. A precision of 93.7% indicates a low false-alarm rate, an important property for a system intended to

trigger corrective AV responses. Early Fusion V2 does exhibit rapid overfitting after epoch 2, attributable to only ~18 unique non-empty BEV configurations in the 57-clip training set; the epoch-2 checkpoint was retained via validation-based early stopping, and its result was independently verified on the untouched test episodes.

4.2. Time-to-Collision Safety Analysis

As F1 weights every frame in a labelled event window equally regardless of physical imminence, a kinematic time-to-collision (TTC) measure was computed per test frame from relative position and closing speed, capped at 30 s and flagged as safety-critical below 3 s [5], [6]. Both detected and missed events span the full TTC range, while missed detections (false negatives) show a higher median TTC (~6 s) than false positives (~3–4 s), indicating that Early Fusion V2’s errors skew toward less kinematically acute moments. Raising the decision threshold increases mean TTC at detection only marginally (from 24 s to 28 s) while reducing true positives by roughly a third, supporting the use of an F1-optimized rather than a safety-margin-maximized threshold.

5. Discussion and Conclusion

These results show that kinematic motion data alone is a remarkably strong predictor of third-party rash behavior, that naively appended visual/LiDAR features are not unconditionally beneficial, and that preserving spatial structure in the LiDAR representation, via a CNN-processed BEV grid rather than scalar statistics, is necessary for multi-modal fusion to exceed the kinematic-only baseline under a genuine episode-level generalization test. Achieving this entirely on CPU hardware, without a dedicated GPU, supports deployability on constrained onboard AV compute platforms such as NVIDIA Drive or Mobileye EyeQ.

This dissertation’s contributions include a reactive threat-anticipation task framing distinct from ego-centric aggressive driving detection; a purpose-built MetaDrive dataset with frame-precise multi-modal labels; the Early Fusion V2 architecture, achieving 0.9003 test F1 on unseen road environments; and a demonstration that this result is achievable on CPU-only hardware. Limitations include the modest dataset scale (50 episodes), the simulation-only setting, a binary label that does not distinguish cut-in from braking events, and a scope restricted to vehicle-to-vehicle interaction. Future work will pursue sim-to-real transfer, per-event-type classification, a longer prediction horizon, and expanded dataset scale and BEV spatial diversity to further improve generalization.

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